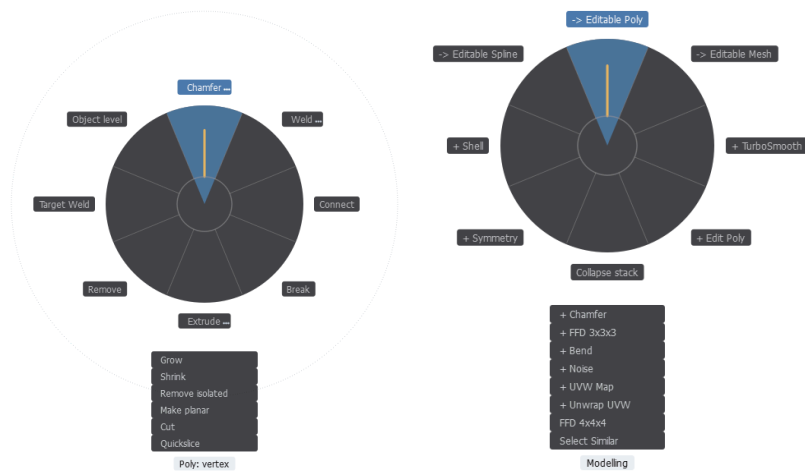


MarkingForge

Marking menus, hotbox and gesture sliders for Autodesk 3ds Max 2027

INSTALLATION AND CONFIGURATION GUIDE · VERSION 1.4.3



REAL MARKING MENUS - A QUICK GESTURE NEEDS NO DIAL

MarkingForge works the way marking menus are meant to: the command is chosen by the **direction** of the movement, not by clicking an item. Press the key and flick the mouse towards the command in one quick movement - the command runs and **the dial is not even drawn**. The dial appears only when you hold the key and wait, to remind you where things are. After a few days your hand knows the directions and you stop looking.

Hold a key, flick the mouse, done. This guide covers installing the plugin, setting up shortcuts, using the 24 built-in dials, changing them, and adding MarkingForge commands to 3ds Max's menus, toolbars and quad menus. Every option in detail, with tips and troubleshooting, is in the companion document **MarkingForge Reference**.

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1. What you get

The download is one ZIP file. Unpacked, it holds:

MarkingForge_1.4.3\ MarkingForge_1.4.3.mzp	installer - drag onto a viewport (method 1)
Documentation\ MarkingForge_Installation_Guide.pdf	this document
MarkingForge_QuickStart.pdf	tutorials: first gestures, the editor
MarkingForge_Reference.pdf	every option, tips, troubleshooting
MarkingForge_Shortcut_Card.pdf	the 24 dials on one page to print
MarkingForge_for_Maya_Users.pdf	Maya habits mapped to MarkingForge
MarkingForge_Preset_Packs.pdf	the four ready-made dial sets
MarkingForge_Studio_Deployment.pdf	silent install for studios
Presets*.json	Modelling, Animation, Look-dev, Archviz
Studio\Deploy-MarkingForge.ps1	silent install / update / removal
README.txt	the short version of this document
CHANGELOG.txt	what is in this version
LICENSE.txt	licence agreement
Manual install\ MarkingForge_Install.ms	the same installer as a script (method 2)
MarkingForge\ PackageContents.xml	the plugin folder itself (method 3)
plugins\MarkingForge.gup	tells 3ds Max what to load
scripts\MarkingForge_Startup.ms	the plugin (C++)
macroscripts\MarkingForge_Macros.mcr	builds the MarkingForge menu
tools\mf_editor*.py	the menu commands
Documentation*.pdf	the menu editor (Python)
	these guides - MarkingForge > Documentation

What kind of plugin is it?

MarkingForge is a **native C++ plugin** - a Global Utility Plugin, the `.gup` file, which is a Windows DLL that 3ds Max loads when it starts. The dials, the gesture recognition and the drawing all run in C++ with Qt, the same toolkit 3ds Max itself is built with, so a dial appears in a few milliseconds.

Around it sit three small parts: a **startup script** (MAXScript) that adds the MarkingForge menu to the menu bar, a set of **macroscripts** (MAXScript) behind the menu commands - which is also what lets you put them on toolbars and quad menus - and the **menu editor**, written in Python with PySide6, both of which ship with 3ds Max 2027. Nothing else needs installing.

Everything is installed as one folder that 3ds Max loads through its standard *Application Plugins* mechanism. Nothing is copied into the 3ds Max program folder, and nothing is installed outside your user profile unless you choose "All users".

2. Requirements

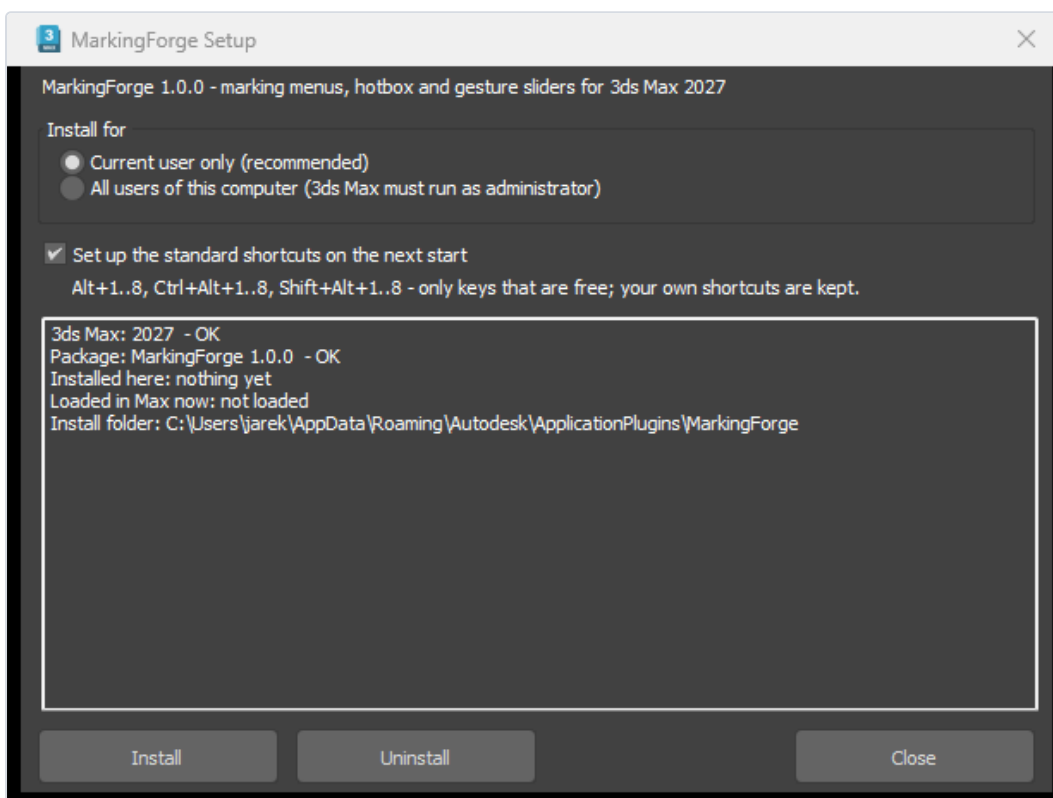
Item	Requirement
3ds Max	Autodesk 3ds Max 2027, Windows 64-bit. This download is for 3ds Max 2027 only - it does not load in other versions, and the installer checks and refuses. MarkingForge is available for 3ds Max 2025-2027, each version with its own download; 3ds Max 2024 and earlier may follow if enough users ask for them.
Windows	Windows 10 or 11, 64-bit.
Visual C++ runtime	Microsoft Visual C++ 2015-2022 Redistributable (x64). 3ds Max installs it itself, so it is almost always present. If the plugin does not load, reinstall it from Microsoft (<code>vc_redist.x64.exe</code>).
Rights	None special for "Current user". "All users" needs 3ds Max started as administrator.

3. Installation - three methods

All three methods end in the same place: the folder `MarkingForge` in `%APPDATA%\Autodesk\ApplicationPlugins\`. Pick the one you like; methods 1 and 2 run the same installer.

Method 1 - drag and drop the .mzp RECOMMENDED

- 1 Unzip the download and start 3ds Max 2027.
- 2 Drag `MarkingForge_1.4.3.mzp` from Explorer and drop it onto any viewport.
- 3 The **MarkingForge Setup** window opens. Leave "Current user only" selected and press **Install**.
- 4 Restart 3ds Max.



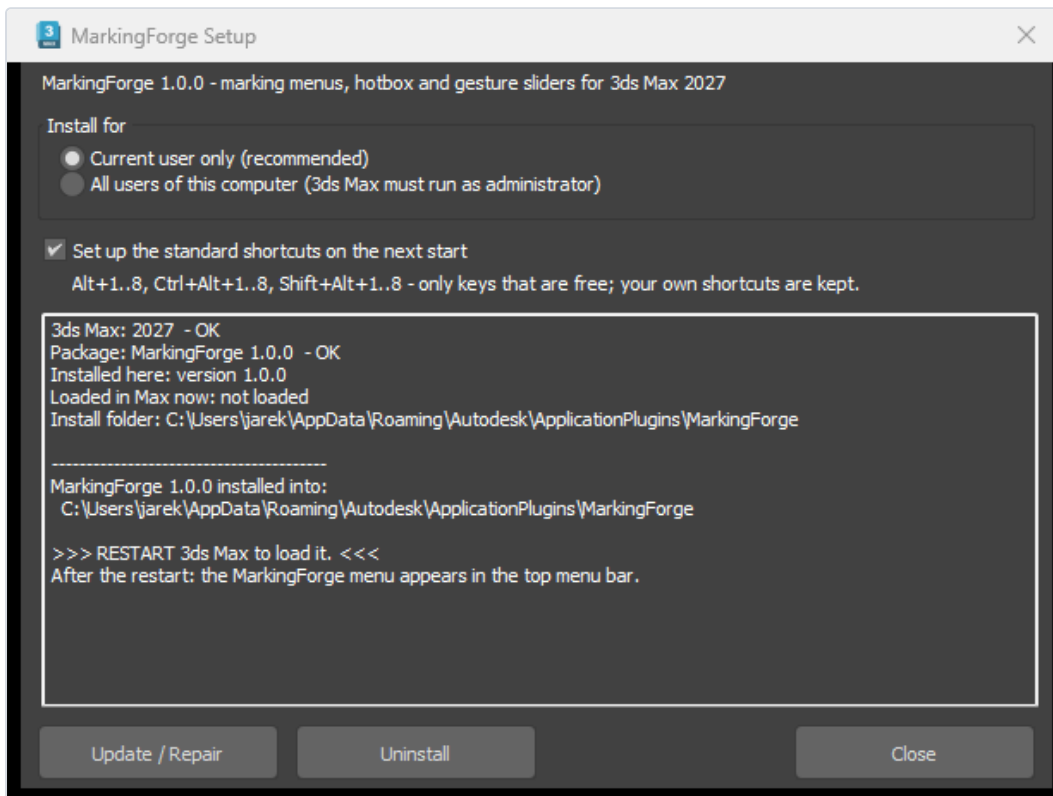
The MarkingForge Setup window. It checks the 3ds Max version, the package and any other copy of the plugin before it lets you install.

Method 2 - run the installer script

- 1 Unzip the download.
- 2 In 3ds Max choose **Scripting > Run Script...**
- 3 Open `Manual\install\MarkingForge_Install.ms`.
- 4 The same Setup window opens - press **Install**, then restart 3ds Max.

Method 3 - copy the folder yourself

- 1 Close 3ds Max.
- 2 Open Explorer, paste `%APPDATA%\Autodesk\ApplicationPlugins` into the address bar and press Enter (create the `ApplicationPlugins` folder if it does not exist).
- 3 Copy the folder `Manual install\MarkingForge` into it.
- 4 Start 3ds Max.



After Install: the Setup window says where the plugin went and asks for a restart.

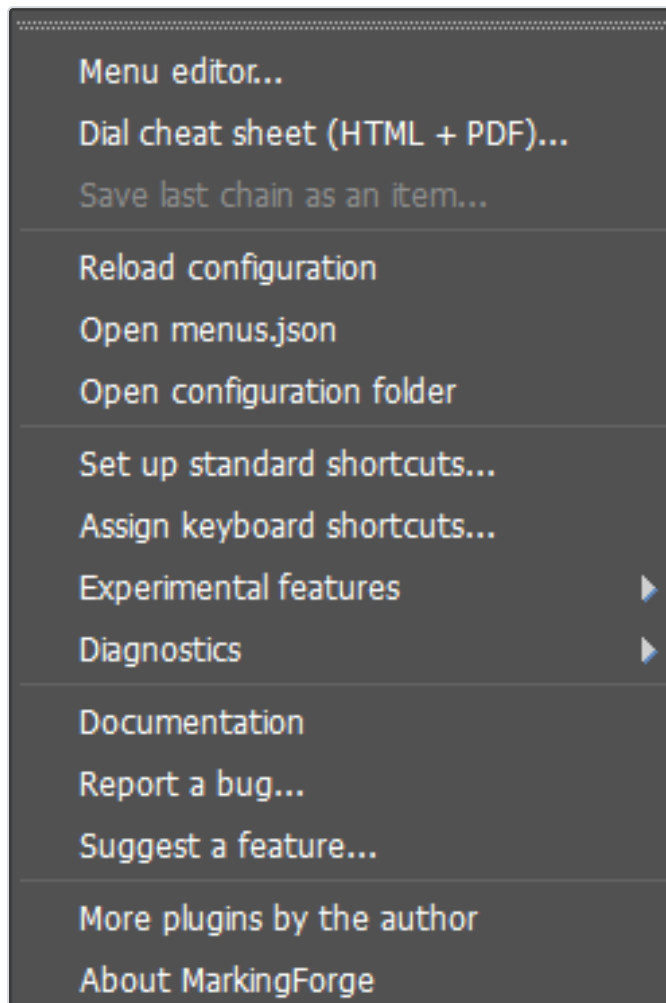
Setup window options

Option	What it does
Current user only	Installs into <code>%APPDATA%\Autodesk\ApplicationPlugins\MarkingForge</code> . No administrator rights needed.
All users of this computer	Installs into <code>%ProgramData%\Autodesk\ApplicationPlugins\MarkingForge</code> . Start 3ds Max as administrator first.
Set up the standard shortcuts on the next start	On the first start after installing, every dial gets its key - Alt+1..8, Ctrl+Alt+1..8, Shift+Alt+1..8. A key you already use for something else is skipped, and no shortcut of yours is changed. Cleared, the first start leaves your keys alone.
Install / Update / Repair	Copies the plugin. Over an existing installation it replaces it, keeping your dials and settings. It works even while the old version is loaded.
Uninstall	Removes the plugin folder. Your own settings are kept (see chapter 12).

Why a restart? MarkingForge is a C++ plugin, and 3ds Max loads C++ plugins only while it starts. The installer copies the files; the next start loads them.

4. Checking that it works

After the restart a **MarkingForge** menu appears in the menu bar, before *Help*.

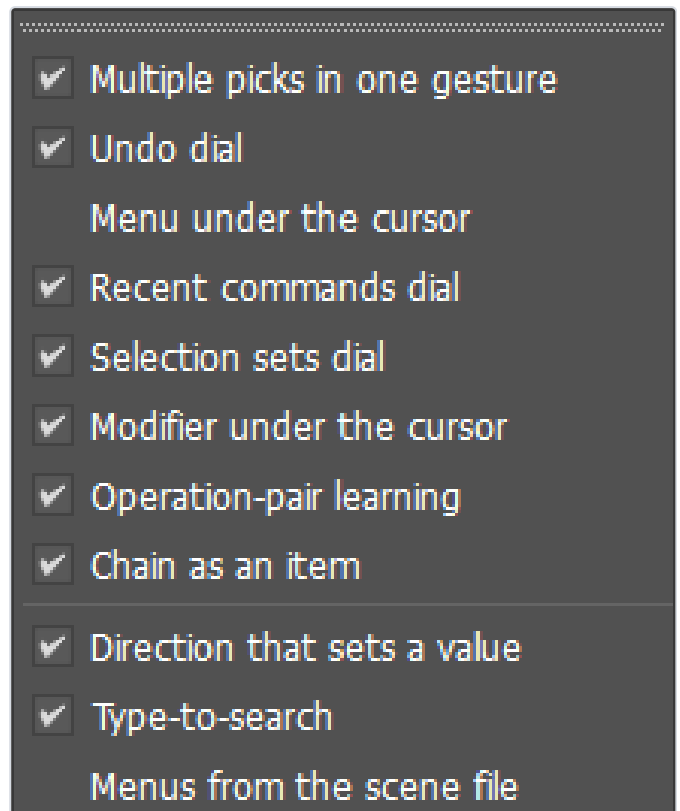


The MarkingForge menu.

To be certain, open the MAXScript Listener (F11) and type:

```
MarkingForge.version()
```

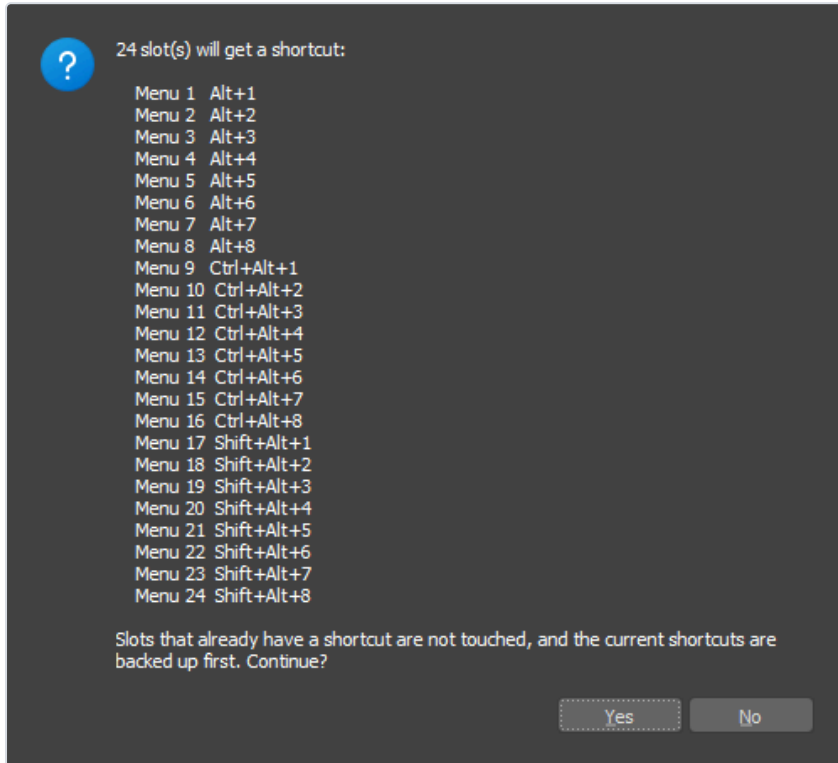
It prints the version number. "Unknown property" means the plugin was not loaded - see Troubleshooting.



Its Experimental features submenu - options you switch on deliberately.

5. Keyboard shortcuts

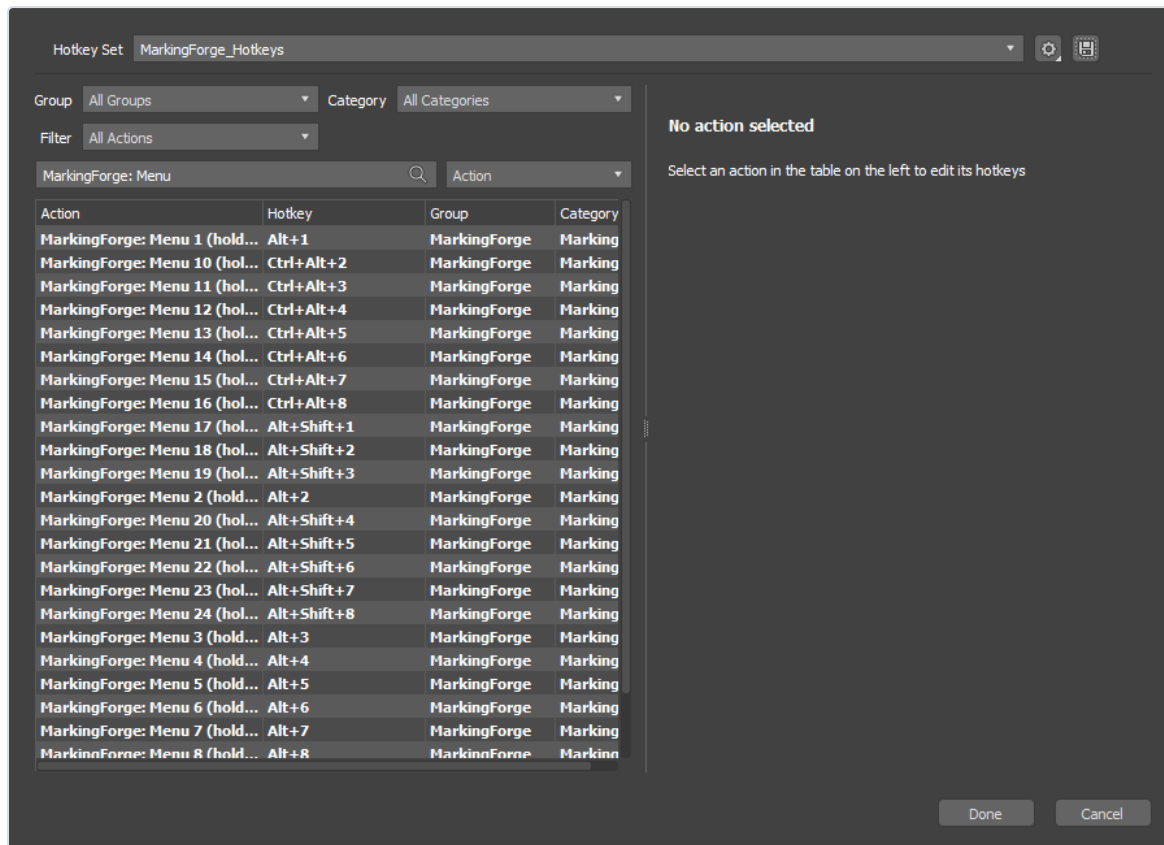
Each of the 24 dials opens by holding a key, and the keys are there from the first start: when no dial has a key yet, the first start gives each its standard one - whatever way MarkingForge was installed - skipping keys something else uses. To fill in dials that are still without a key, choose **MarkingForge > Set up standard shortcuts...** It shows exactly what it will do and asks once:



Set up standard shortcuts: only dials without a shortcut, only keys nothing else uses.

Key	Dial	Key	Dial	Key	Dial
Alt+1	Modelling	Ctrl+Alt+1	Transform	Shift+Alt+1	Materials
Alt+2	Hotbox - Max's menu	Ctrl+Alt+2	Modifiers	Shift+Alt+2	Animation
Alt+3	Create	Ctrl+Alt+3	Show and hide	Shift+Alt+3	Scene
Alt+4	View	Ctrl+Alt+4	Viewports	Shift+Alt+4	File
Alt+5	Selection	Ctrl+Alt+5	Viewport display	Shift+Alt+5	Selection sets
Alt+6	Modifier stack	Ctrl+Alt+6	Snaps and grid	Shift+Alt+6	UV mapping
Alt+7	Undo	Ctrl+Alt+7	Align and pivot	Shift+Alt+7	Tools and setup
Alt+8	Recent commands	Ctrl+Alt+8	Render	Shift+Alt+8	Lights and cameras

You can change any of them in the menu editor ("This slot's shortcut") or in 3ds Max's own **Customize > Hotkey Editor**, where they are listed as *MarkingForge: Menu 1 ... Menu 24*:



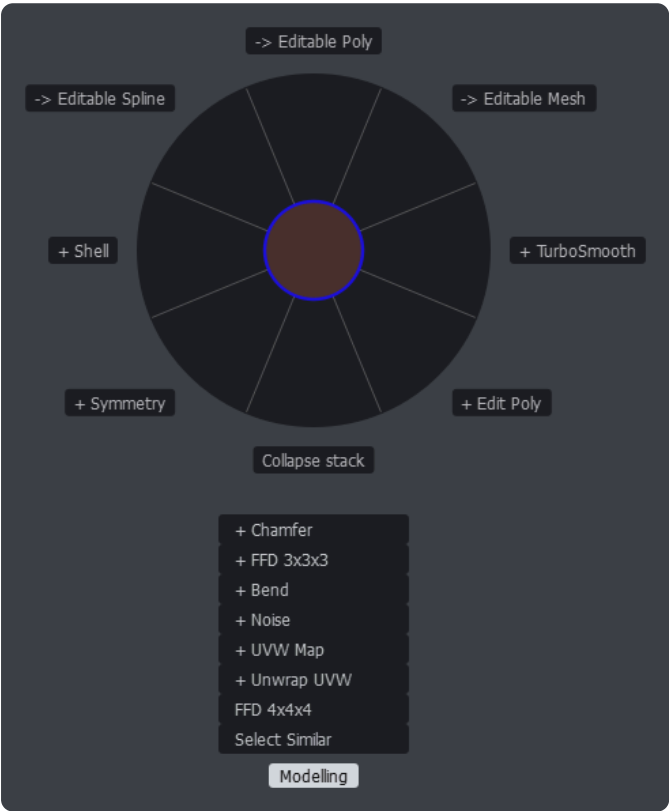
The Hotkey Editor filtered to MarkingForge. Any key can be reassigned here as well.

6. Using the dials

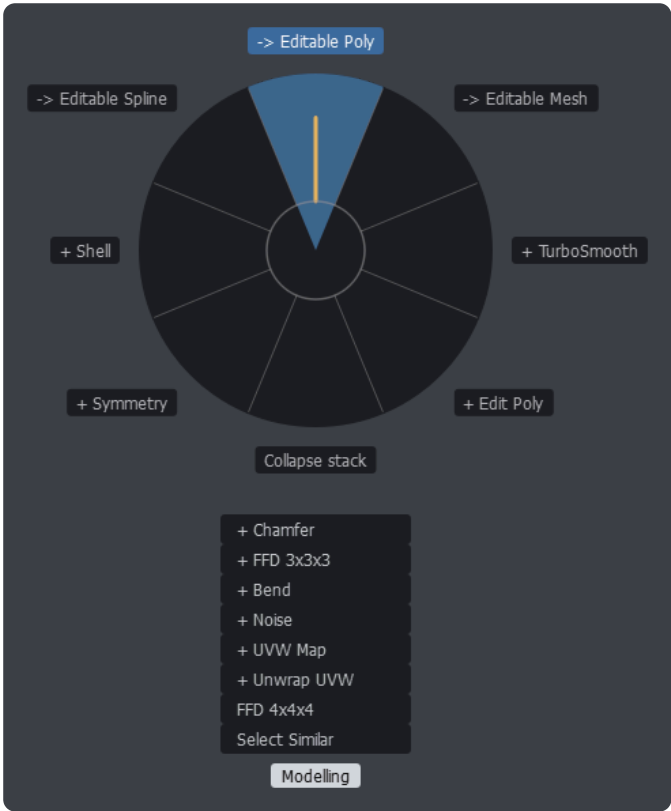
The same key does three things, depending on the hand:

Move	What happens
Hold	Hold the key: the dial appears around the cursor. Move towards an item and release the key. For learning and browsing.
Flick	Press the key and move the mouse at once, without waiting. The item runs without the dial being drawn. After a week or two you stop looking - the direction is in your hand.
Tap	Press and release without moving. Does nothing by default; the editor can give it one of the directions.

Releasing in the centre cancels. So do **Esc** and the right mouse button.



Alt+1 on an ordinary object: convert and add modifiers. Less common items sit in a list under the ring - rows are clicked, directions are aimed at.

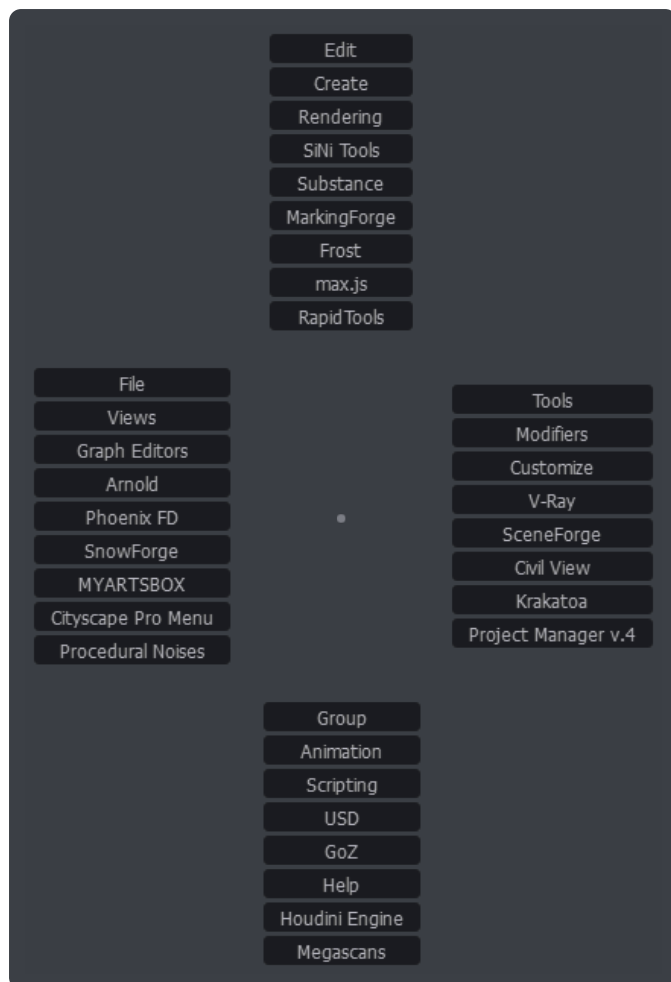


Aiming north: the direction lights up. Releasing runs *Convert to Editable Poly*.

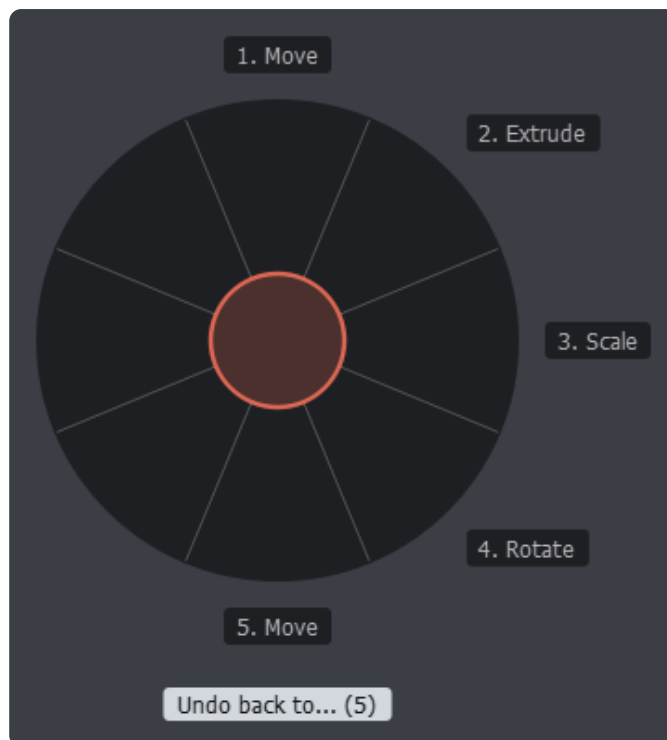
Commands with a settings dialog (the caddy) - Chamfer, Extrude, Inset... - open it on an ordinary release, like the settings icon in a quad menu. Moving **far past the ring** before releasing runs them **without** the dialog - "far" is the **dotted circle** outside the dial, which brightens once you are past it. Either can be made the default per command (**Settings...** in the editor).

More tricks - several commands in one gesture, the dial of the object under the cursor, the suggested next command and search by typing - are in the **QuickStart** ("Five tricks worth knowing") and in chapter 2 of the **Reference**.

Alt+2 is the **hotbox**: all of 3ds Max's main menu as tiles around the cursor, including menus other plugins add - point at a menu title, then at an entry, and release; or click the entry with the left mouse button, as in Maya (a right click closes the hotbox). **Alt+7** is the **undo dial**: the last eight operations by name; N undoes one step, NE two, and so on.



Alt+2 - the hotbox.



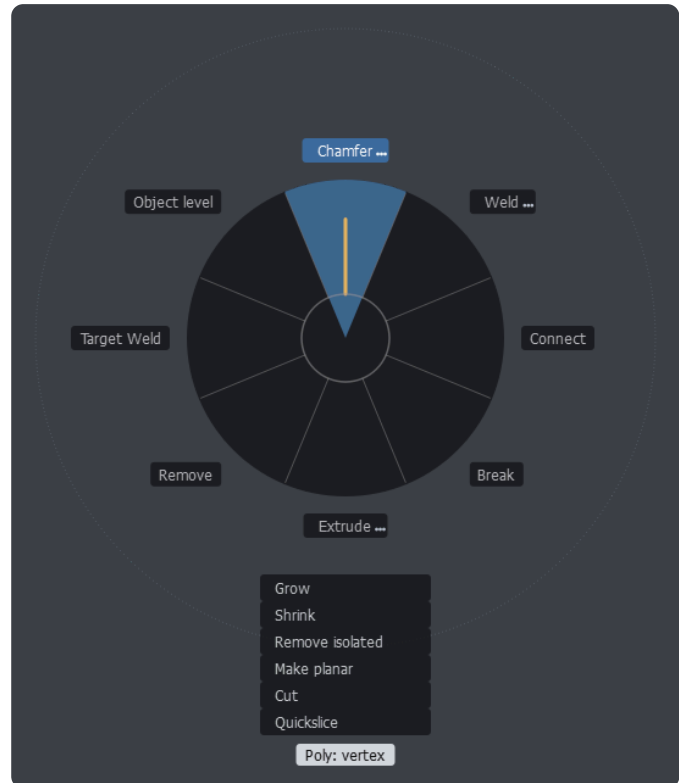
Alt+7 - the undo dial.

7. Dials that follow what you are doing

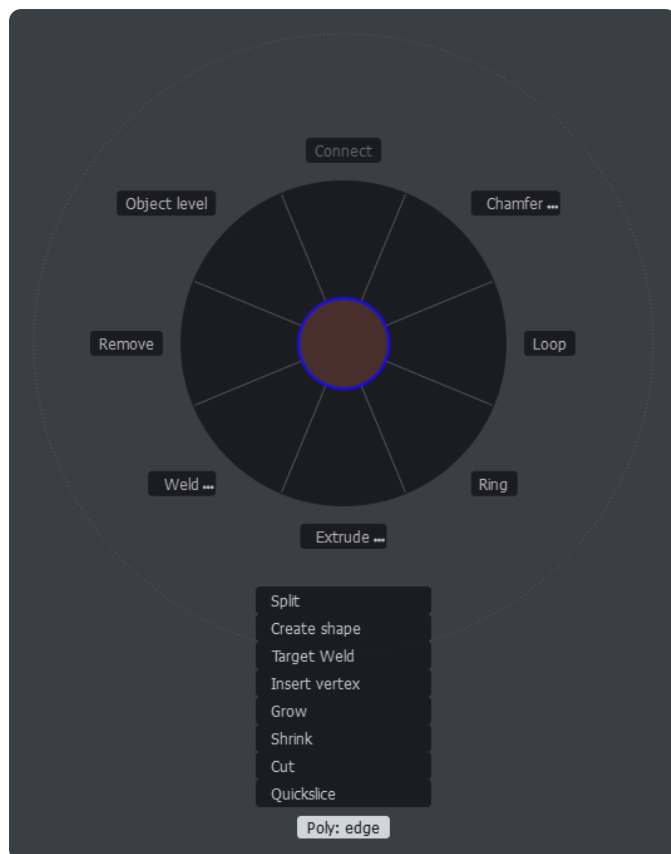
A dial can change its contents with the situation. The built-in **Modelling** dial (Alt+1) is the clearest example: *Convert to Editable Poly* and everything you do next live under the same key.



Alt+1 on an Editable Poly at object level: the sub-object levels, Attach, TurboSmooth, Collapse.



The same key at Vertex level: Chamfer, Weld, Connect, Break... NW always returns to object level.



Edge level: Connect, Chamfer, Loop and Ring opposite each other.

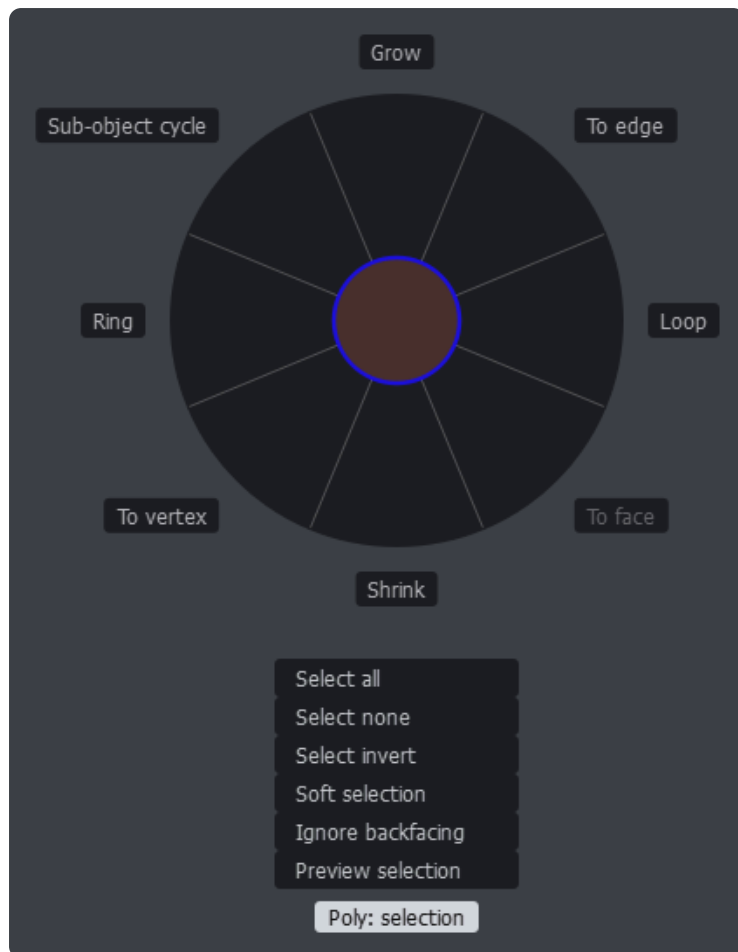


Polygon level, aiming at Inset.

The **same operation keeps the same direction** on an Editable Poly and under an Edit Poly modifier, so your hand does not have to know which one it is working on:

Level	N	NE	E	SE	S	SW	W	NW
Object	Vertex	Edge	Border	Polygon	Element	Attach	+ TurboSmooth	Collapse stack
Vertex	Chamfer	Weld	Connect	Break	Extrude	Remove	Target Weld	Object level
Edge	Connect	Chamfer	Loop	Ring	Extrude	Weld	Remove	Object level
Border	Cap	Chamfer	Extrude	Connect	Create shape	Shrink	Grow	Object level
Polygon	Extrude	Bevel	Inset	Outline	Hinge	Flip	Detach	Object level
Element	Detach	Hide	Flip	Unhide all	Hide unselected	Shrink	Grow	Object level

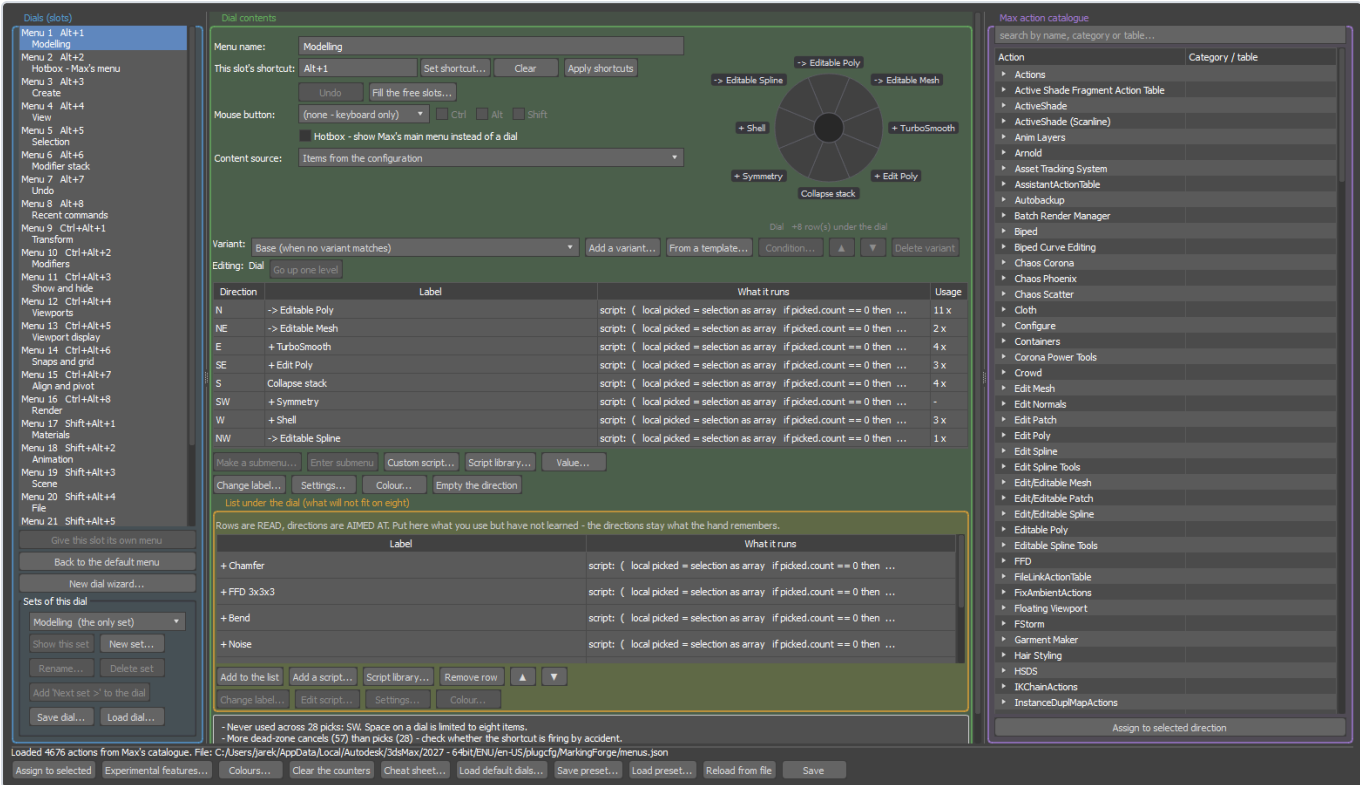
Other dials do the same: Selection offers Grow, Shrink, Loop and Ring at sub-object level; Create turns into shapes when a shape is selected; Lights and cameras turns into camera or light controls; Materials, Animation and Scene change when the cursor is over the Slate editor, Track View or the Scene Explorer. Chapter 11 lists every case.



Alt+5 on an Editable Poly sub-object level: Grow and Shrink opposite each other, Loop and Ring opposite each other.

8. The menu editor

MarkingForge > Menu editor... Everything about the dials is set here; changes take effect as soon as you press **Save**, without restarting 3ds Max.



The menu editor: dials on the left, the selected dial's directions in the middle, 3ds Max's 4000+ commands on the right.

Area	Use
Dials (left)	The 24 slots with their shortcuts. Pick one to edit it.
Shortcut and mouse button	The key that opens the dial; optionally a mouse button with a modifier (for example Ctrl+Shift+Middle).
Content source	Items from the configuration, or a live source: modifier stack, undo list, recent commands, selection sets.
Variant	The contextual variants of the dial and their conditions (object class, sub-object level, modifier, window under the cursor, selection count, name, layer).
Directions	N ... NW. Drag a command from the catalogue onto a row, or double-click a command. Custom script, Value (a gesture slider), Submenu and Colour are the buttons below.
List under the dial	Up to twelve extra rows for what does not fit in eight directions.
Catalogue (right)	Every command 3ds Max knows, including other plugins' - searchable by name.
Bottom row	Assign to selected, Experimental features, Colours, Clear the counters, Cheat sheet (a printable PDF of your saved dials), Load default dials, Save / Load preset (share your layout), Reload from file, Save. The buttons wrap to more rows when the window is narrow.

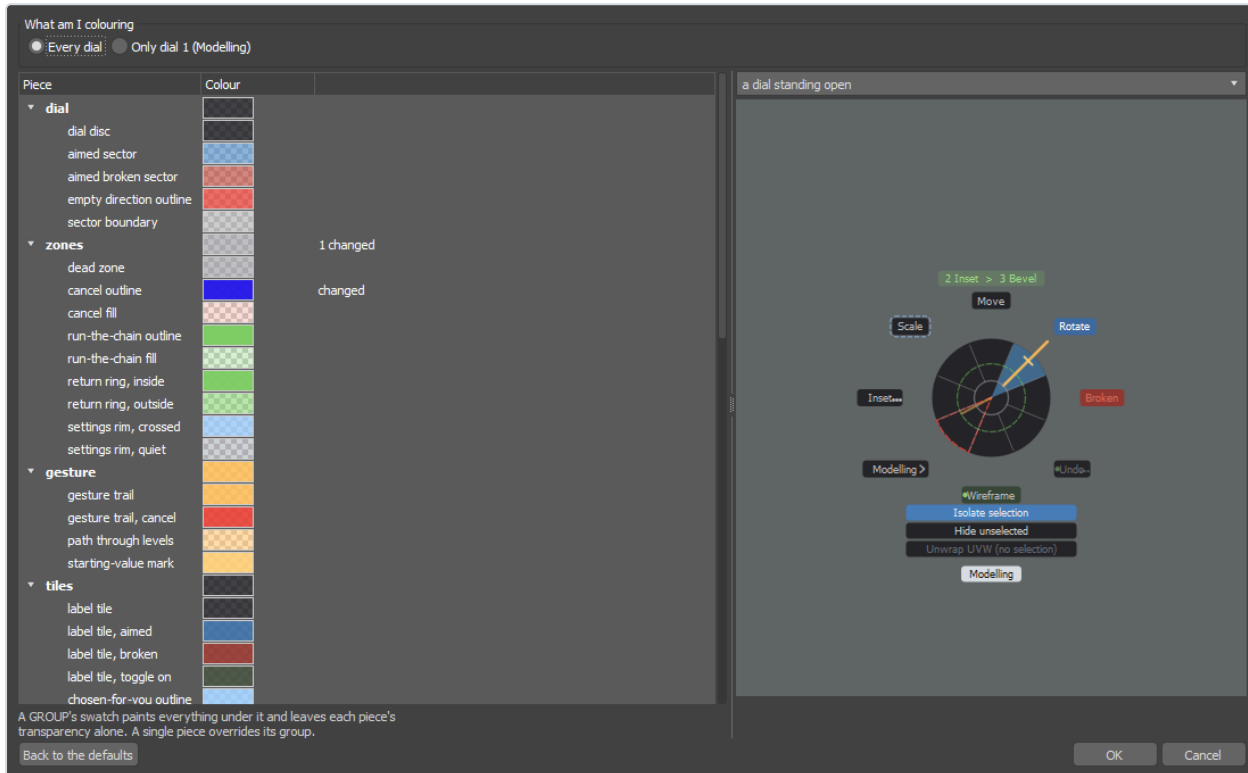
To get the built-in layout back at any time: **Load default dials...** By default it fills only the dials that have no menu of their own.

Closing the editor with unsaved changes (Esc or the window's X) asks whether to **Save**, **Discard** or **Cancel** - work is never thrown away without a question. If `menus.json` changed on disk while the editor was open, Save does not overwrite it: it offers **Save your edits as a separate copy**.

9. Colours, mouse buttons and experimental features

Colours

Colours... in the editor sets the palette for all dials or for one dial only; **Colour...** under the directions colours a single item. The preview shows the dial at rest, aimed, cancelling and chaining.



The colour editor.

Opening a dial with a mouse button

Next to the shortcut, pick a mouse button and at least one modifier key. A bare button is refused on purpose: left selects, middle pans and right opens the quad menu - taking one of those away would break 3ds Max.

Experimental features

Options that change how gestures behave - several picks in one gesture, the menu of the object under the cursor, type-to-search, gesture sliders. Switch them in **Experimental features...** in the editor or in the MarkingForge menu.

These features WORK, but each has a reason for not being on by default. The reason sits under the name - read it first. Switching them all off restores the behaviour from before they existed.

☒ **Direction that sets a value (drag further = more)**

Instead of firing an action, the direction lets you keep dragging, and the distance from the centre sets a number. The release commits it.

WHY EXPERIMENTAL: it needs the current value read from the scene, and that cost has not yet been measured on the path to the first pixel.

☒ **Typing with the menu open searches for actions**

You hold the key, start typing, and the dial shows the best matches from Max's action catalogue.

WHY EXPERIMENTAL: the overlay deliberately does NOT take the keyboard and does not take the focus - working over Slate ME and Track View rests on that. This feature touches that foundation.

☐ **Menus loaded from the scene file**

A menu can travel with a scene - useful when a rigger hands over a scene with a ready-made menu for their rig.

WHY EXPERIMENTAL: somebody else's scene can carry 'script' items, which run with FULL rights. Switch this on only for scenes you trust.

☒ **Multiple picks in one gesture**

Cross the ring and come back to the dashed circle round the centre - the item waits in a queue. Releasing in that circle (it is green then) runs the whole queue as ONE undo.

To give up: release at the very CENTRE (the small dot) or over an EMPTY direction - the queue is discarded.

WHY EXPERIMENTAL AND OFF: it gives meaning to a movement that used to mean nothing - going a little past the ring and coming back is also how a hand gives up on a direction. Operation order can matter too - though a wrong order undoes with a single Ctrl+Z.

☒ **Undo dial**

Offers the 'Undo stack' content source: the last eight operations BY NAME. Direction N undoes one, NE two, and so on.

To use it, set Source = Undo stack on some slot and give that slot a shortcut.

WHY EXPERIMENTAL: undoing several steps at once is irreversible other than through Redo, and the operation names are read via MAXScript, because the SDK does not expose the undo stack.

☐ **Menu under the cursor**

The dial describes the object UNDER THE CURSOR rather than the selected one. Picking an item selects it and runs the operation on it - in one undo.

An ALREADY SELECTED object does not change the selection, so a multiple selection stays whole and hovering over any one of its objects acts on the whole group.

WHY EXPERIMENTAL: the hit test walks the scene. Dials whose contents depend on the object run it before they appear; the others appear first and name the object a moment later. If it takes over 8 ms three times in a row, it switches itself off for the rest of the session and says so in the diagnostics.

☒ **Recent commands dial**

Offers the 'Recent commands' content source: the eight most recently RUN commands, newest first. For REPEATING, as opposed to the undo stack.

Repeating a command moves it to the front, so what you use most stays under your hand by itself.

The history lives in MEMORY and starts afresh when Max starts - it describes what you are doing now, not what you did last week.

WHY EXPERIMENTAL: a record is written after EVERY action that runs. The cost is negligible (one insertion into an eight-element buffer, off the road to the first pixel), but it is a new step in a place that everything passes through.

☒ **Selection sets dial**

Offers the 'Selection sets' content source: the scene's named sets as eight directions. With more than eight it shows the first eight.

WHY EXPERIMENTAL: the sets are read straight from the SDK, but their

OK

Cancel

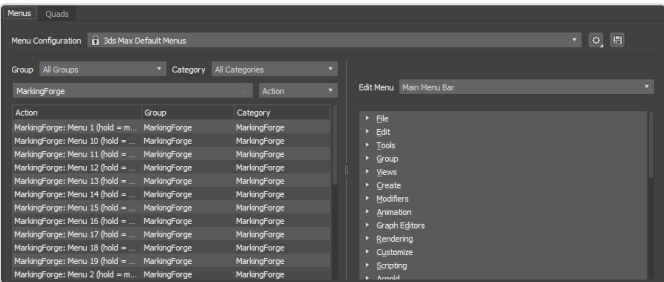
The experimental features, each with the reason it is optional.

10. Putting MarkingForge into the 3ds Max interface

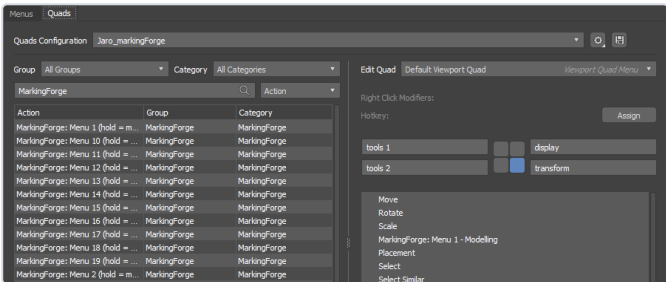
Every MarkingForge command is a standard 3ds Max action in the category **MarkingForge**. You can put any of them on a toolbar, into a menu or a quad menu, or give it a hotkey - exactly like 3ds Max's own commands.

Into a menu or a quad menu

- 1 Choose **Customize > Menu Editor**.
- 2 On the **Menus** tab (or **Quads** for the right-click quad menu) type **MarkingForge** in the action search field.
- 3 Drag an action from the list into the menu tree on the right.
- 4 Close the editor with **Done**.



Menu Editor - MarkingForge actions ready to drag into a menu.



The same actions on the Quads tab, for the right-click quad menu.

Onto a toolbar

Choose **Customize > Customize User Interface**, open the **Toolbars** tab, pick the category **MarkingForge** and drag an action onto any toolbar. A button shows the command's name; hovering it shows the description below.

The commands

Command	What it does
Menu editor...	Opens a window where you assign actions to dial directions and set keyboard shortcuts.
Dial cheat sheet (HTML + PDF)...	Collects every dial into one document to read and to print. Along the way it checks which dial would open empty.
Save last chain as an item...	Takes the last chain you ran (multiple picks) and stores it as ONE direction in a dial of your choice.
Reload configuration	Reads menus.json again. Useful when the file was edited outside Max.
Open menus.json	Opens the configuration file in your default text editor. The plugin picks up changes by itself.
Open configuration folder	Opens the folder holding menus.json, the shortcut backups and the settings in Explorer.
Assign keyboard shortcuts...	Opens 3ds Max's Hotkey Editor. The dials are the actions 'MarkingForge: Menu 1' ... 'Menu 24' - type MarkingForge into its search field.
Set up standard shortcuts...	Gives every dial without a shortcut its standard one: Alt+1..8, Ctrl+Alt+1..8, Shift+Alt+1..8. Keys already in use are skipped; your own shortcuts are not touched.
Configuration status	Shows which file is loaded, how many menus it holds, whether it has errors, and the state of the experimental features (including how often pair-learning suggestions were taken).

About MarkingForge	Shows the plugin's version, its configuration file, and links to support and to more plugins by the author.
Documentation	Opens the folder with the MarkingForge guides (QuickStart, Reference, Installation and more).
Report a bug...	Copies your 3ds Max and MarkingForge details to the clipboard and opens an e-mail to forgeplugins@gmail.com .
Suggest a feature...	Send an idea or a feature request to forgeplugins@gmail.com .
More plugins by the author	Opens https://forgeplugins.tech in your web browser.
Multiple picks in one gesture	Cross the ring and come back to the centre - the item waits in a queue. The release runs the whole queue as ONE undo.
Undo dial	Offers the "undo" menu source: the last eight operations by name. Assign it to a slot in the menu editor.
Menu under the cursor	The dial describes the object UNDER THE CURSOR rather than the selected one. Picking an item selects that object and runs the operation on it - in a single undo.
Direction that sets a value	A "value" direction turns into a slider - the gesture radius sets the number.
Type-to-search	Typing while the dial is open turns it into a search over Max's actions.
Menus from the scene file	CAUTION: menus carried in by somebody else's scene. Script items are rejected, but switch this on deliberately all the same.
Modifier under the cursor	A dial with the 'Modifier stack' source shows the stack of the object UNDER THE CURSOR rather than the selected one. Requires 'Menu under the cursor' - without it there is nowhere to get the object from.
Operation-pair learning	The plugin remembers what follows what and sets the dial's DEFAULT direction to the most frequent successor. It does NOT move items around - muscle memory rests on that. Session memory only.
Chain as an item	After you run a chain (multiple picks) the plugin remembers it. You then store it with 'Save last chain...' - with no dialog popping up after every gesture.
Recent commands dial	Offers the 'Recent commands' content source: the last eight you ran, newest first. For REPEATING, as opposed to the undo stack.
Selection sets dial	Offers the 'Selection sets' content source: the scene's named sets as eight directions.
Menu 1 - next set ... Menu 24 - next set	Switches that dial to its next set - one action per dial, for a key of its own. Sets are made in the menu editor.

The 24 dials themselves are the actions *MarkingForge: Menu 1 ... Menu 24*. They are meant for keys and mouse buttons: a dial lives while the key is held, so a toolbar button cannot hold it open.

11. The 24 built-in dials

The layout a fresh installation starts with. Opposite directions hold opposite operations (All / None, Grow / Shrink, Hide / Unhide), so the hand learns one movement and its reverse. The grids show the directions as they sit on screen; the dial's name is in the centre.

Alt+1

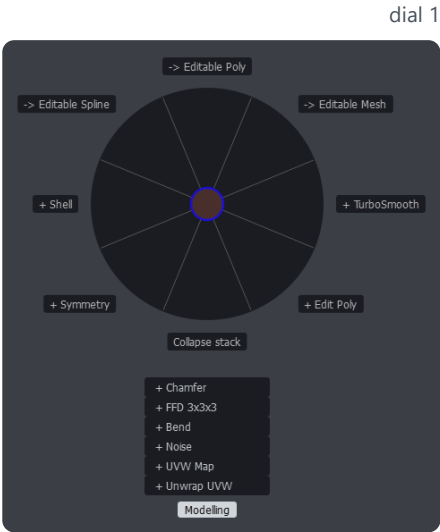
Modelling

NW -> Editable Spline	N -> Editable Poly	NE -> Editable Mesh
W + Shell	Modelling	E + TurboSmooth
SW + Symmetry	S Collapse stack	SE + Edit Poly

List under the dial: + Chamfer · + FFD 3x3x3 · + Bend · + Noise · + UVW Map · + Unwrap UVW

Changes with the situation (first match wins):

- **Materials (Slate)** - when cursor over the Slate Material Editor
- **Track View** - when cursor over Track View
- **Particle View** - when cursor over Particle View
- **Scene Explorer** - when cursor over the Scene Explorer
- **Modifier stack** - when cursor over the command panel
- **UV (Unwrap)** - when Unwrap UVW is current in the Modify panel
- **Edit Poly: vertex** - when Edit Poly is current in the Modify panel, sub-object level: Vertex
- **Edit Poly: edge** - when Edit Poly is current in the Modify panel, sub-object level: Edge
- **Edit Poly: border** - when Edit Poly is current in the Modify panel, sub-object level: Border / Segment
- **Edit Poly: polygon** - when Edit Poly is current in the Modify panel, sub-object level: Polygon
- **Edit Poly: element** - when Edit Poly is current in the Modify panel, sub-object level: Element
- **Edit Poly: object** - when Edit Poly is current in the Modify panel
- **Poly: vertex** - when the object is an Editable Poly, sub-object level: Vertex
- **Poly: edge** - when the object is an Editable Poly, sub-object level: Edge
- **Poly: border** - when the object is an Editable Poly, sub-object level: Border / Segment
- **Poly: polygon** - when the object is an Editable Poly, sub-object level: Polygon
- **Poly: element** - when the object is an Editable Poly, sub-object level: Element
- **Poly: object** - when the object is an Editable Poly
- **Spline: vertex** - when a shape (spline) is selected, sub-object level: Vertex
- **Spline: spline** - when a shape (spline) is selected, sub-object level: Spline
- **Spline** - when a shape (spline) is selected
- **Nothing selected** - when nothing is selected



Alt+2 Hotbox - Max's menu

Shows tiles built from Max's whole main menu - including menus other plugins add. Point at the entry and release, or click it; your selection is not changed.

dial 2



Alt+3 Create

NW Cameras and lights >	N Box	NE Sphere
W Helpers >	Create	E Cylinder
SW More primitives >	S Shapes >	SE Plane

Submenu Shapes (S): N Line · NE Arc · E Circle · SE Ellipse · S Rectangle · SW NGon · W Helix · NW Text

Submenu More primitives (SW): N Cone · NE Torus · E Tube · SE GeoSphere · S Pyramid · SW Teapot · W Chamfer Box · NW Capsule

Submenu Helpers (W): N Dummy · NE Point · E Grid · SE Tape · S Protractor · SW Compass · W Expose Transform · NW Container

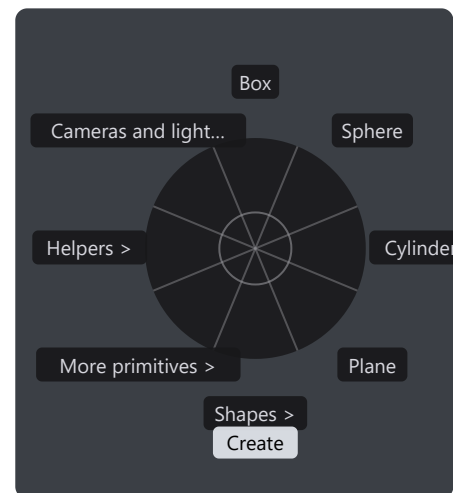
Submenu Cameras and lights (NW): N Target camera · NE Physical camera · E Free camera · SE Sun positioner · S Free light · SW Target light · W Omni light · NW Match camera to view

List under the dial: Star · Donut · Hedra · Oil Tank

Changes with the situation (first match wins):

- **Create shapes** - when a shape (spline) is selected

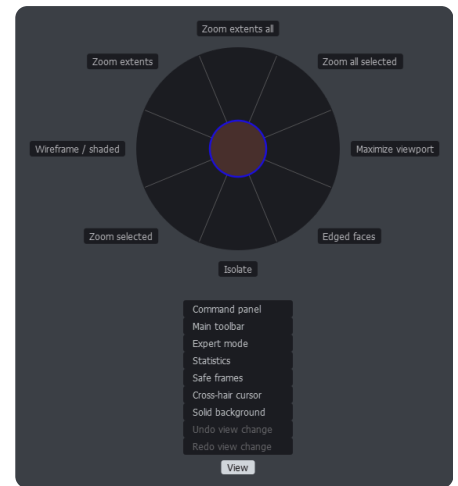
dial 3



Alt+4 View

NW Zoom extents	N Zoom extents all	NE Zoom all selected
W Wireframe / shaded	View	E Maximize viewport
SW Zoom selected	S Isolate	SE Edged faces

List under the dial: Command panel · Main toolbar · Expert mode · Statistics · Safe frames · Cross-hair cursor · Solid background · Undo view change · Redo view change



dial 4

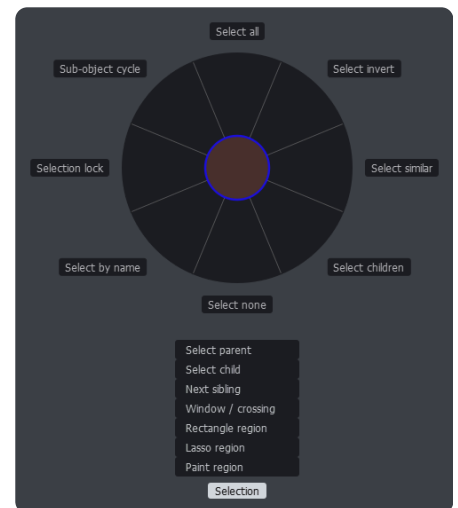
Alt+5 Selection

NW Sub-object cycle	N Select all	NE Select invert
W Selection lock	Selection	E Select similar
SW Select by name	S Select none	SE Select children

List under the dial: Select parent · Select child · Next sibling · Window / crossing · Rectangle region · Lasso region · Paint region

Changes with the situation (first match wins):

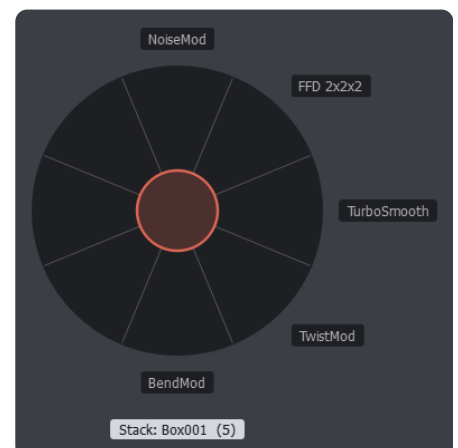
- **Selection** - when Edit Poly is current in the Modify panel, sub-object level: object level
- **Edit Poly: selection** - when Edit Poly is current in the Modify panel
- **Selection** - when the object is an Editable Poly, sub-object level: object level
- **Poly: selection** - when the object is an Editable Poly



dial 5

Alt+6 Modifier stack

Filled live from the modifier stack of the selected object (top eight).

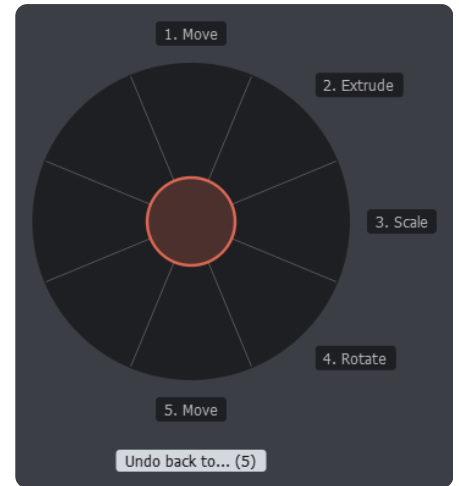


dial 6

Alt+7 Undo

Filled live from Max's undo list - N undoes one step, NE two, and so on.

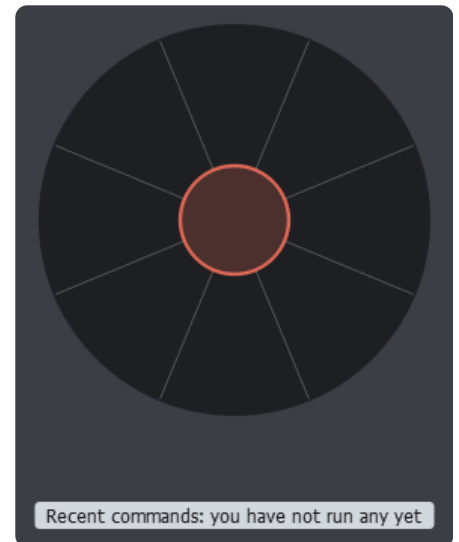
dial 7



Alt+8 Recent commands

Filled live from the eight commands you ran most recently.

dial 8

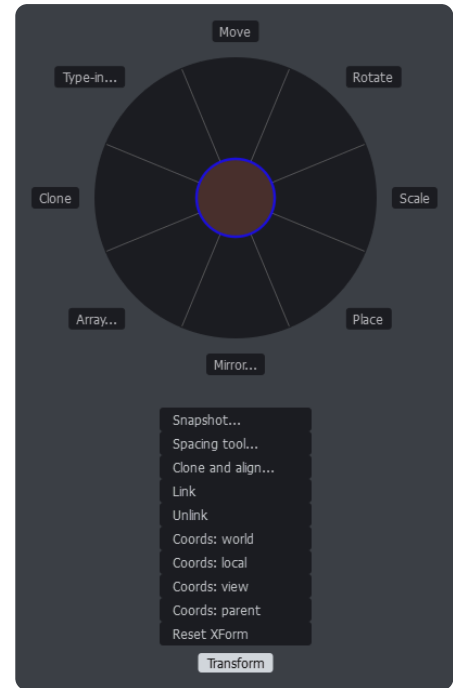


Ctrl+Alt+1 Transform

dial 9

NW Type-in...	N Move	NE Rotate
W Clone	Transform	E Scale
SW Array...	S Mirror...	SE Place

List under the dial: Snapshot... · Spacing tool... · Clone and align... · Link · Unlink · Coords: world · Coords: local · Coords: view · Coords: parent · Reset XForm



Ctrl+Alt+2 Modifiers

dial 10

NW Geometry >	N TurboSmooth	NE Symmetry
W Slice	Modifiers	E Shell
SW Deform >	S + Edit Poly	SE Chamfer

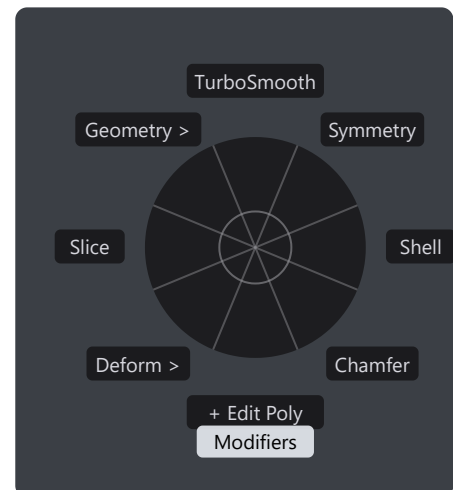
Submenu Deform (SW): N + Bend · NE + Twist · E + Taper · SE + Noise · S + FFD 3x3x3 · SW + Relax · W + Stretch · NW + Skew

Submenu Geometry (NW): N Mirror · NE Array · E Cap Holes · SE Vertex Weld · S Face Extrude · SW + Quadify Mesh · W OpenSubdiv · NW MeshSmooth

List under the dial: + Unwrap UVW · + UVW Map · + Turn to Poly · + ProOptimizer · + Material ID

Changes with the situation (first match wins):

- **Shape modifiers** - when a shape (spline) is selected



Ctrl+Alt+3 Show and hide

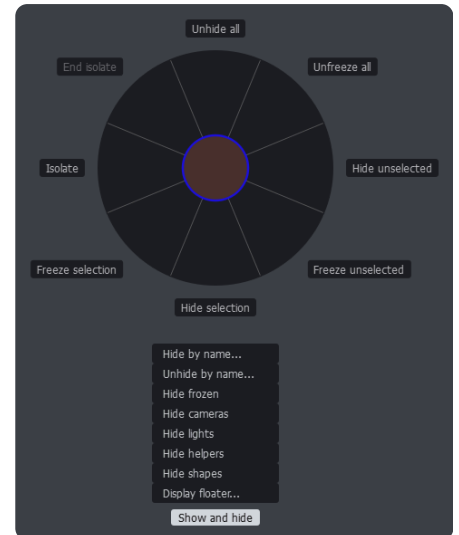
dial 11

NW End isolate	N Unhide all	NE Unfreeze all
W Isolate	Show and hide	E Hide unselected
SW Freeze selection	S Hide selection	SE Freeze unselected

List under the dial: Hide by name... · Unhide by name... · Hide frozen · Hide cameras · Hide lights · Hide helpers · Hide shapes · Display floater...

Changes with the situation (first match wins):

- **Show and hide** - when the object is an Editable Poly, sub-object level: object level
- **Poly: show and hide** - when the object is an Editable Poly

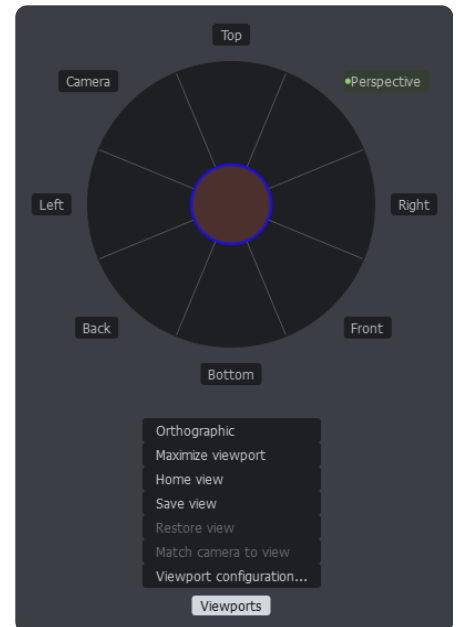


Ctrl+Alt+4 Viewports

dial 12

NW Camera	N Top	NE Perspective
W Left	Viewports	E Right
SW Back	S Bottom	SE Front

List under the dial: Orthographic · Maximize viewport · Home view · Save view · Restore view · Match camera to view · Viewport configuration...

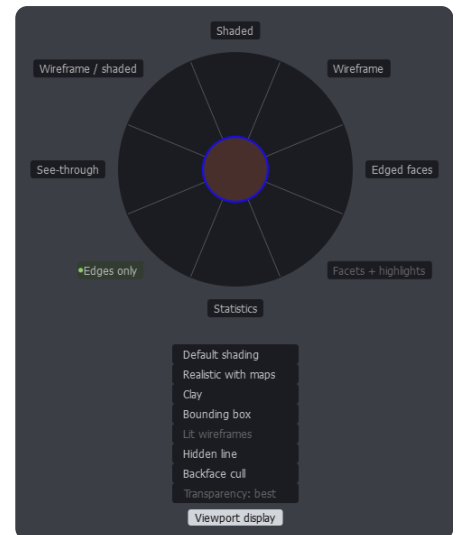


Ctrl+Alt+5 Viewport display

dial 13

NW Wireframe / shaded	N Shaded	NE Wireframe
W See-through	Viewport display	E Edged faces
SW Edges only	S Statistics	SE Facets + highlights

List under the dial: Default shading · Realistic with maps · Clay · Bounding box · Lit wireframes · Hidden line · Backface cull · Transparency: best

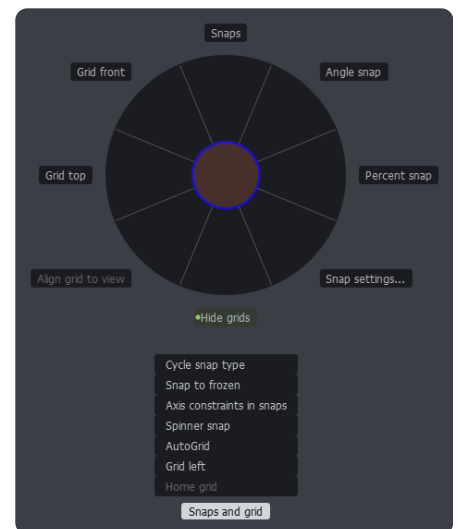


Ctrl+Alt+6 Snaps and grid

dial 14

NW Grid front	N Snaps	NE Angle snap
W Grid top	Snaps and grid	E Percent snap
SW Align grid to view	S Hide grids	SE Snap settings...

List under the dial: Cycle snap type · Snap to frozen · Axis constraints in snaps · Spinner snap · AutoGrid · Grid left · Home grid



Ctrl+Alt+7 Align and pivot

dial 15

NW Working pivot to view	N Quick align	NE Normal align...
W Pin working pivot	Align and pivot	E Align to view...
SW Place pivot surface	S Edit working pivot	SE Align camera

List under the dial: Align... · Place pivot view · Use working pivot · Reset working pivot · Center pivot · Pivot to origin · Reset XForm

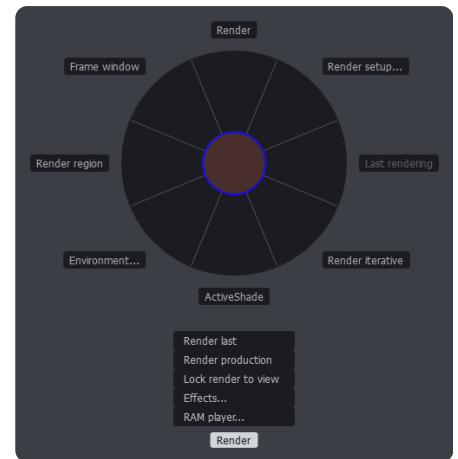


Ctrl+Alt+8 Render

dial 16

NW Frame window	N Render	NE Render setup...
W Render region	Render	E Last rendering
SW Environment...	S ActiveShade	SE Render iterative

List under the dial: Render last · Render production · Lock render to view · Effects... · RAM player...



Shift+Alt+1 Materials

dial 17

NW Shaded with maps	N Slate editor	NE Compact editor
W Realistic with maps	Materials	E Material/Map Browser
SW + UVW Map	S Edit selected's material	SE Material Explorer

List under the dial: Condense slots · Reset slots · Restore slots · + Material ID

Changes with the situation (first match wins):

- **Materials (Slate)** - when cursor over the Slate Material Editor



Shift+Alt+2 Animation

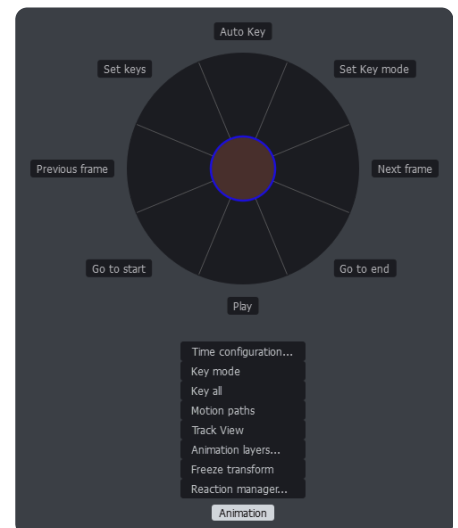
dial 18

NW Set keys	N Auto Key	NE Set Key mode
W Previous frame	Animation	E Next frame
SW Go to start	S Play	SE Go to end

List under the dial: Time configuration... · Key mode · Key all · Motion paths · Track View · Animation layers... · Freeze transform · Reaction manager...

Changes with the situation (first match wins):

- **Track View** - when cursor over Track View



Shift+Alt+3 Scene

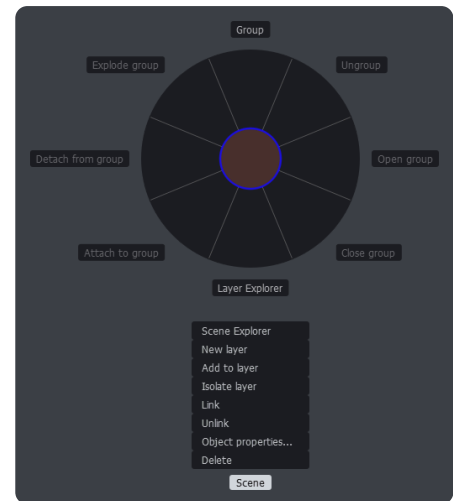
dial 19

NW Explode group	N Group	NE Ungroup
W Detach from group	Scene	E Open group
SW Attach to group	S Layer Explorer	SE Close group

List under the dial: Scene Explorer · New layer · Add to layer · Isolate layer · Link · Unlink · Object properties... · Delete

Changes with the situation (first match wins):

- **Scene Explorer** - when cursor over the Scene Explorer

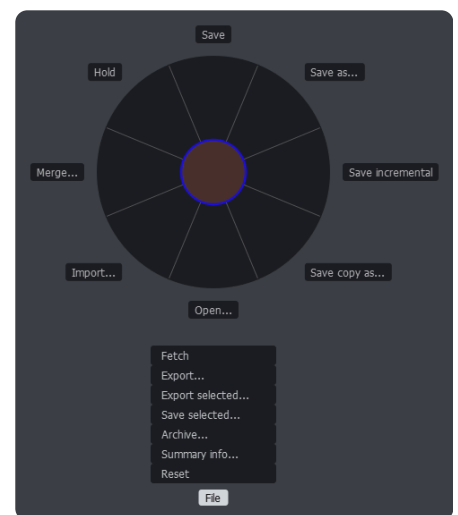


Shift+Alt+4 File

dial 20

NW Hold	N Save	NE Save as...
W Merge...	File	E Save incremental
SW Import...	S Open...	SE Save copy as...

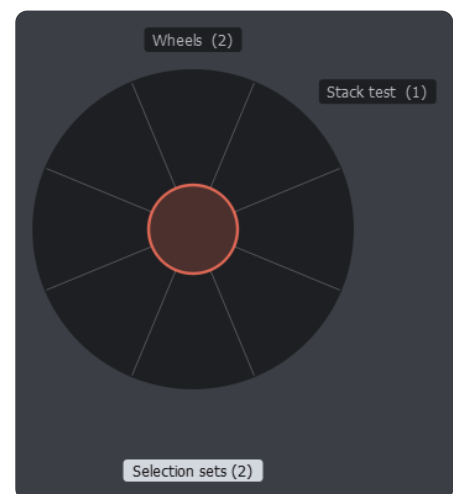
List under the dial: Fetch · Export... · Export selected... · Save selected... · Archive... · Summary info... · Reset



Shift+Alt+5 Selection sets

dial 21

Filled live from the named selection sets of the scene.



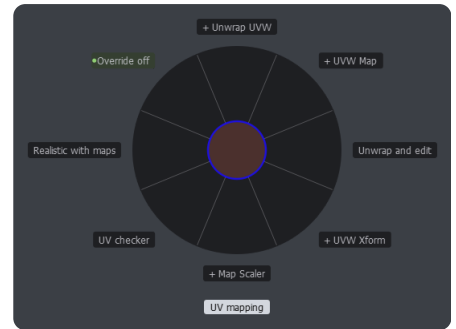
Shift+Alt+6 UV mapping

dial 22

NW Override off	N + Unwrap UVW	NE + UVW Map
W Realistic with maps	UV mapping	E Unwrap and edit
SW UV checker	S + Map Scaler	SE + UVW Xform

Changes with the situation (first match wins):

- **UV (Unwrap)** - when Unwrap UVW is current in the Modify panel

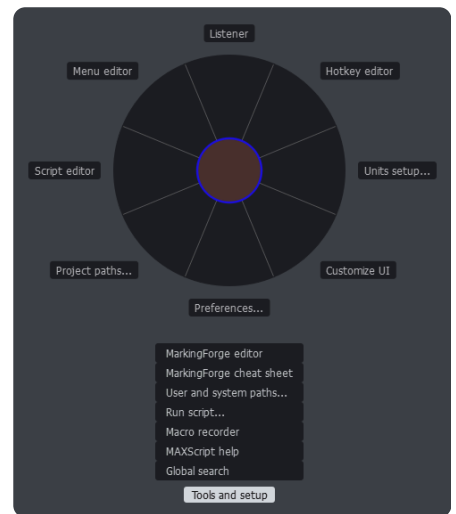


Shift+Alt+7 Tools and setup

dial 23

NW Menu editor	N Listener	NE Hotkey editor
W Script editor	Tools and setup	E Units setup...
SW Project paths...	S Preferences...	SE Customize UI

List under the dial: MarkingForge editor · MarkingForge cheat sheet · User and system paths... · Run script... · Macro recorder · MAXScript help · Global search



Shift+Alt+8 Lights and cameras

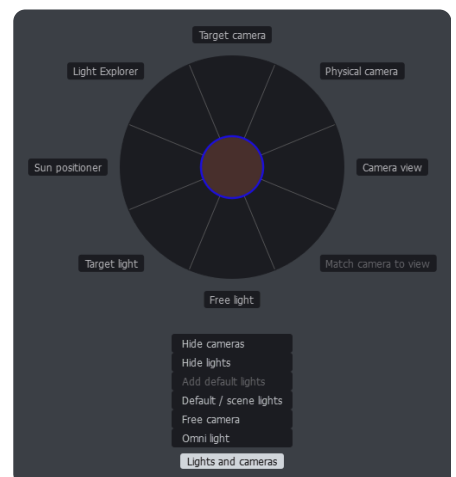
dial 24

NW Light Explorer	N Target camera	NE Physical camera
W Sun positioner	Lights and cameras	E Camera view
SW Target light	S Free light	SE Match camera to view

List under the dial: Hide cameras · Hide lights · Add default lights · Default / scene lights · Free camera · Omni light

Changes with the situation (first match wins):

- **Camera** - when a camera is selected
- **Light** - when a light is selected



12. Files and settings

What	Where
The plugin	%APPDATA%\Autodesk\ApplicationPlugins\MarkingForge
Your dials	%LOCALAPPDATA%\Autodesk\3dsMax\2027 - 64bit\ENU\en-US\plugcfg\MarkingForge\menus.json
Shortcut backups	the same folder, hotkeys\backup_*.hsx - one before every change
Usage counters, logs	the same folder

MarkingForge > Open configuration folder opens it in Explorer; **Open menus.json** opens the file in your text editor. The plugin notices a saved file and reloads it by itself.

To move your setup to another computer, copy menus.json - or use **Save preset...** in the editor, which leaves out paths and usage data.

13. Updating and uninstalling

Updating

Install the new version the same way as the first time. The Setup window shows **Update / Repair**; your dials and shortcuts are kept. Restart 3ds Max.

An update cannot leave you without a working plugin: the installer first backs up the installed version (into 3ds Max's temp folder), and if copying the new one fails - a full disk, an antivirus lock - it puts the previous version back and says so. Nothing to do but fix the cause and run it again.

Uninstalling

- 1 Run the installer (method 1 or 2) and press **Uninstall**, then restart 3ds Max - or close 3ds Max and delete the plugin folder from chapter 12.
- 2 Optional: delete the settings folder `plugcfg\MarkingForge` as well - but NOT its `hotkeys` subfolder: 3ds Max uses the shortcut file in it as your active shortcut set, with ALL your shortcuts, not only MarkingForge's. Assign another set in **Customize > Hotkey Editor** first if you want that folder gone too.

14. Troubleshooting

Symptom	Cause and fix
No MarkingForge menu after the restart	Type <code>MarkingForge.version()</code> in the Listener. "Unknown property" = not loaded. Check that the folder is in <code>ApplicationPlugins</code> (chapter 12) and contains <code>PackageContents.xml</code> ; reinstall the Visual C++ 2015-2022 x64 runtime.
3ds Max reports a duplicate plugin at start	Another copy of <code>MarkingForge.gup</code> exists, usually in the 3ds Max <code>plugins</code> folder. The Setup window lists every copy it finds - delete all but the one in <code>ApplicationPlugins</code> .
A key does not open a dial	Check it in Customize > Hotkey Editor : another command may use the same key. Run Set up standard shortcuts... or set the key in the menu editor.
One shortcut never reaches 3ds Max (e.g. Ctrl+Alt+5), or pressing it brings up another program	Another program has taken that key combination for the whole of Windows, and Windows gives it to that program before 3ds Max sees it - no setting inside 3ds Max can change that. Common owners: PowerToys (ZoomIt uses Ctrl+Alt+digits), graphics-driver overlays (NVIDIA, AMD, Intel), Discord, screenshot tools, mouse and keyboard software (Logitech, Razer). Find the program by pressing the key while 3ds Max is in front and watching which window appears, then change or switch off the shortcut in that program - or give the dial another key in the menu editor.
A dial opens empty	It uses a live source with nothing to show (no modifiers, no selection sets). The cheat sheet lists such dials at the top.
Save says the configuration changed after it was opened	<code>menus.json</code> was changed on disk while the editor was open (another editor window, or a tick in <code>MarkingForge > Experimental features</code>). Save your edits as a separate copy, press Reload from file and carry the edits over.
The dial shows a different menu than expected	A contextual variant matched. <code>MarkingForge.eventLog()</code> names the variant that won.
The editor does not open	The error is shown in a message; also check that the <code>tools</code> folder is inside the plugin folder.
A <code>menus.json</code> edited by hand stopped working	MarkingForge > Diagnostics > Configuration status names the line with the error. A broken item disables only that item.

Listener commands

```
MarkingForge.version()      -- is the plugin loaded
MarkingForge.configSummary() -- errors and warnings in menus.json
MarkingForge.eventLog()     -- the last gestures: what opened, what ran
MarkingForge.windowInfo()   -- what the plugin sees under the cursor
MarkingForge.mouseInfo()    -- mouse-button bindings
```

MarkingForge sends nothing over the network, does not touch the keyboard outside 3ds Max and changes your shortcuts only when you ask it to.

Still stuck? Open an issue at github.com/looki666/MarkingForge/issues or write to forgeplugins@gmail.com.
MarkingForge > Report a bug... puts your 3ds Max and MarkingForge details on the clipboard, ready to paste.

15. Support, bug reports and feature requests

ISSUES	github.com/looki666/MarkingForge/issues
E-MAIL	forgeplugins@gmail.com
WEB	https://forgeplugins.tech (forgeplugins.tech - more plugins by the author)

Bug reports and feature requests are best opened as issues on GitHub - the forms there ask for exactly what is needed, and you can follow the fix. Questions, and anything you would rather not post publicly (licence, purchase), go to the e-mail address. Every message is read.

Reporting a bug

- 1 In 3ds Max choose **MarkingForge > Report a bug...** - it copies your 3ds Max version, MarkingForge version, configuration summary and start-up log to the clipboard. It works even when the plugin did not load.
- 2 Open github.com/looki666/MarkingForge/issues and pick **Bug report** - or answer **Yes** in the dialog to write an e-mail to forgeplugins@gmail.com instead.
- 3 Say what you did, what you expected and what happened instead, then paste the details with **Ctrl+V**.
- 4 Add a screenshot if you can. When the problem is in one dial, attach a preset of it (editor > **Save preset...**).

Suggesting a feature

Open an issue with the **Feature request** form, choose **MarkingForge > Suggest a feature...**, or simply write. Describe what you would like to do and how you do it today - a concrete example from your work says more than a list of wishes. Ideas for new built-in dials, commands and options are all welcome.

Nothing is sent by the plugin itself: the details go to the clipboard and your own e-mail program, so you see everything that leaves your computer. MarkingForge has no network access.